Zoology Legends

BT403

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Join our Legend Group

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Team Zoology LEGENDS

Monitoring the Growth of Culture

To monitor the growth following should be checked:

- Cell number
- PCV
- Fresh wt. and Dry wt.
- Cell Viability
- Medium conductivity & pH
- Use two methods simultaneously till stationary phase

Advantages of biofertilizers / Role of bio fertilizer

Biofertilizers are known to play a number of vital roles in soil fertility, crop productivity and production in agriculture as they are ecofriendly and cannot at any cost replace chemical fertilizers that are indispensable for getting maximum crop yields.

Some of the important functions or roles of Biofertilizers in agriculture are:

• They supplement chemical fertilizers for meeting the integrated nutrient demand of the crops.

• They can add 20-200 kg N/ha year (eg. Rhizobium sp 50-100 kg N/ha year ; Azospirillum , Azotobacter : 20-40 kg N/ha /yr; Azolla : 40-80 kg N/ha; BGA :20-30 kg N/ha) under optimum

soil conditions and thereby increases 15-25 percent of total crop yield.

• They can at best minimize the use of chemical fertilizers not exceeding 40-50 kg N/ha under ideal agronomic and pest-free conditions.

• Application of Biofertilizers results in increased mineral and water uptake, root development, vegetative growth and nitrogen fixation.

• Some Biofertilizers (eg, Rhizobium BGA, Azotobacter sp) stimulate production of growth promoting substance like vitamin-B complex, Indole acetic acid (IAA) and Gibberellic acids etc.

Advantages and disadvantage of herbicide

Advantages of using herbicides

• Broad spectrum of weeds controlled

• Reduced crop injury

• Reduced herbicide carryover

• New mode of action for resistance management

• Crop management flexibility and simplicity

• Use of herbicides that are more environmentally friendly

Disadvantages of herbicides

- Mammalian toxicity
- Eco toxicity
- Weeds become super weeds
- Reduced crop yield
- Creates soil and air pollution

Herbicides also damage the Crop plants along with weed

Outcomes of controlled pollination in plants.

Controlled pollination of plants led to the realization that specific crosses could result in a new generation that performed better in the field than either of the parents or the progeny of subsequent generations, i.e. the expression of heterosis through hybrid vigour was observed.

Because one of the two major activities in plant breeding is manipulating genetic variability, a key prerequisite to successful plant breeding is the availability of genetic diversity.

Principle of anther culture medium.

The principle of anther culture medium is to provide a nutrient-rich and controlled environment that supports the growth of haploid plant cells derived from anthers. This medium contains essential nutrients, carbohydrates, growth regulators, vitamins, and a solidifying agent to facilitate pollen germination, cell

division, and embryo development. Sterility is crucial to prevent contamination.

This technique enables the production of haploid plants for various applications in plant breeding and genetic research.

Charcoal affect in the anther and pollen culture

- Stimulates embryogenesis in anther cultures of plants like potato, rye, and tobacco.
- Removes inhibitory substances from agar, enhancing growth.
- Absorbs degradation products of sucrose, like 5-2-furfural.
- Eliminates both exogenous (from outside) and endogenous (from within) growth hormones from the culture medium.
- Helps prevent microbial contamination.
- Promotes cell growth and pollen viability in anther and pollen culture.
- Essential for successful plant breeding and genetic studies.

Micro propagation applications:

• Suitable alternative to traditional methods

• High prevalence of plants

• The production of disease-free plants

• Seed production in some crops

• Cost effective process

• Automated micro propagation

• Very small size explants can be used

• Only practical method of multiplying genetically modified cells or cells after protoplast fusion.

• Ease in keeping, packing and transport of material multiplied by micro propagation

How Plants Respond to Disease:

Defense compounds: Plants produce chemicals to fight diseases, acting as natural pesticides.

Immune gene activation: Genes activate to trigger defense responses against pathogens.

Isolation of infected areas: Plants isolate infected tissues to prevent pathogen spread and minimize damage.

Define transformation with example and how plants are transformed!

Transformation is when a cell changes genetically because it takes in genes from outside its own body. This usually happens in bacteria, but it can also occur naturally in some other living things.

Example:

A common example of plant transformation is the insertion of genes encoding for insect resistance into crops like corn or cotton. By introducing genes from bacteria or other sources that produce insecticidal proteins, plants can defend themselves against insect pests without the need for external pesticide applications.

How plants are transformed:

Plant transformation involves inserting DNA from one organism into the genome of a plant to give it a new trait. This added DNA is called a "transgene," making the plant "transgenic."

The process includes:

- Introducing the desired gene into plant cells.
- Integrating the introduced DNA into the plant's genome.
- Culturing transformed cells to grow into complete plants.
- Identifying transformed cells using selective media.
- Confirming the presence and expression of the desired trait in transgenic plants.

Why plant virus use a gene transgene virus

Plant viruses utilize gene transgene to manipulate host cells, facilitating their own replication and spread. These transferred genes modify cellular processes, create favorable conditions,

and evade host defenses, ensuring the virus's survival and propagation.

How synthetic seeds are produced

Synthetic seed can be defined as the artificial encapsulation of somatic embryo, shoot, bud, or aggregates of cells or any tissue which has the ability to form a plant in vitro.

Synthetic seeds offer advantages over natural seeds as their advantages in convenient handling, mass production, disease-free propagation, long-term storage, and efficient transfer of desired traits.

Development in plants

• Irreversible change in state.

• It means the qualitative change in plant body.

• e.g. Seed Seedling Vegetative maturation Flowering.

• Growth is a continuous process and Development is phase to phase process

Write principle of meristem culture.

Meristem culture involves aseptically culturing excised shoot tips or meristems on nutrient medium to induce growth. Shoots develop directly or from callus formation, leading to multiple shoots.

These shoots can be further propagated by nodal cuttings, where each segment containing a node grows into another shoot.

Meristem culture is a powerful technique for rapid propagation and genetic preservation of plants, enabling the production of multiple identical shoots from a single meristem. This method is widely used in plant tissue culture for the mass production of disease-free and genetically uniform plantlets

Agrobacterium mediated transformation

Agrobacterium-mediated transformation is a versatile method used in both dicotyledonous and monocotyledonous plants, as well as in yeasts and fungi. The transferred DNA (T-DNA) carried by Agrobacterium can induce insertion mutations, making it useful for mutagenesis studies.

In plants like Arabidopsis thaliana, the method helps find where these mutations happen in specific genes.

Haploid cell

Haploid refers to a cell or organism that contains a single set of chromosomes. In humans, this means having 23 chromosomes, as opposed to the usual diploid number of 46 chromosomes. Haploid cells are typically found in gametes (sperm and egg cells) and certain stages of the life cycle in fungi and algae.

Diploid Cell:

Diploid refers to a cell or organism that contains two sets of chromosomes, one set inherited from each parent. Most somatic cells in organisms, including humans, are diploid. This means they have the full complement of chromosomes, usually denoted as 2n, where 'n' represents the haploid number of chromosomes.

Embryo culture

Embryo culture is a plant biotechnology technique where embryos at different developmental stages are isolated from maternal tissues and grown in vitro in a nutrient-rich medium.

This controlled environment allows the embryos to develop into plantlets, aiding in plant propagation and genetic studies.

Tumor causing part of Agrobacterium tumefaciens and how it can be transformed into plant.

The tumor-causing part of Agrobacterium tumefaciens is a DNA plasmid called the Ti plasmid. In the early 1970s, researchers discovered that this plasmid was responsible for inducing tumors in plants.

By removing the cancer-causing genes from the Ti plasmid and introducing novel genes, researchers were able to exploit A. tumefaciens' natural ability to infect plants. This

process involves allowing the bacteria to insert the desired DNA into the genomes of the plants they infect.

However, not all plant cells are susceptible to infection by A. tumefaciens. To overcome this limitation, alternative methods were developed, including electroporation and micro-injection.

Another significant breakthrough came with the invention of the Biolistic Particle Delivery System, commonly known as the gene gun, by John Sanford in 1990. This technology allows for the direct delivery of DNA into plant cells by shooting DNA-coated particles into plant tissues. This method has proven to be highly effective in transforming plant cells with desired genes, bypassing the need for bacterial infection.

Protoplast and it's methods of isolation.

Protoplasts are naked plant cells without the cell wall, but they possess plasma membrane and all other cellular components.

• They represent the functional plant cells but for the lack of the barrier, cell wall. Protoplasts of different species can be fused to generate a hybrid and this process is referred to as somatic hybridization (or protoplast fusion).

Protoplast isolation

• Refers to the separation of protoplast from plant tissue

• Important to isolate viable and uninjured protoplast as gently and as quickly as possible

• **Involves two methods:**

Mechanical method: This involves physically disrupting the plant tissue to release protoplasts. It may include chopping, grinding, or blending the tissue.

Enzymatic method: This method employs enzymes to break down the cell wall and release protoplasts. Enzymes like cellulose and pectinase are commonly used to digest the cell wall while preserving the integrity of the protoplasts.

Preparation of stock solution for tissue culture technique.

Preparation of stock solutions involves creating concentrated solutions of the necessary components of a medium, which are then diluted as needed. This process is employed to save time and reduce the tedium of weighing individual components each time a medium is prepared.

Key points about stock solutions:

Stock solutions are concentrated solutions containing the essential components of a medium. By diluting these solutions, one can easily prepare the desired medium.

Types of stock solutions:

Simple stock solutions: Comprise only one constituent.

Complex stock solutions: Contain multiple chemicals necessary for the medium.

Components: Stock solutions may include macro and micronutrients, vitamins, and growth regulators.

Quality of water: Stock solutions are typically prepared using distilled or high-purity demineralized water to prevent contamination.

Chemical grade: It's crucial to use chemicals of the highest grade to ensure the accuracy and reliability of the medium preparation.

Cell suspension for explant

Cell suspension cultures refer to a collection of rapidly dividing cells grown in a liquid medium. These cultures grow faster compared to callus cultures and are more amenable to manipulation. They consist of both cell aggregates and dispersed single cells.

These cultures are often used as a source of cells for explant, which are small tissue samples used to initiate new cultures or experiments.

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Genetic markers, it's types and uses.

Definition:

Genetic markers are specific genes or DNA sequences located on chromosomes.

Types:

- Single Nucleotide Polymorphisms (SNPs)
- Minisatellites
- Microsatellites
- Restriction Fragment Length Polymorphisms (RFLPs)

Uses:

- Genetic Mapping: Identifying gene locations on chromosomes.
- Population Genetics: Studying genetic diversity and evolutionary relationships.
- Disease Association Studies: Identifying genes linked to inherited diseases.
- Forensic Identification: DNA profiling for individual identification.
- Breeding: Selecting desirable traits in plants and animals.
- Evolutionary Studies: Tracing evolutionary relationships and processes.

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Write the name of area where protoplast fusion performed.

Protoplast fusion aims to achieve combinations of agriculturally important traits through the creation of somatic hybrids. These hybrids can integrate into conventional breeding programs. Additionally, the extension of protoplast regeneration to a wider range of crop species is crucial for enhancing the applicability of this technique across various agricultural contexts.

Define parasitoids

• Parasitoids are almost the same size as their hosts, and their development always kills the host insect.

• An adult parasitoid deposits one or more eggs into or onto the body of a host insect or somewhere in the host's habitat.

• The larva that hatches from each egg feeds internally or externally on the host's tissues and body fluids, consuming it slowly.

• Later in development, the host dies and the parasitoid pupates inside or outside of the host's body.

• Bathyplectes, trichogramma, encarsia, muscidifurax etc.

Define oxidative stress

Oxidative stress occurs when there's an imbalance between the production of reactive oxygen species (ROS) in the body and the body's ability to get rid of them or repair the damage they cause.

Simply put, it's like having too many harmful molecules and not enough defenses to handle them, leading to damage to cells and tissues.

Environmental factors that cause oxidative stress

• air pollution (increased amounts of ozone or sulfur dioxide)

• oxidant forming herbicides e.g. paraquat dichloride

• heavy metals

• drought

• heat and cold stress

• wounding

• UV light

• intense light that stimulate photo inhibition

Write uses of CaMv in gene transfer.

CaMV, or Cauliflower Mosaic Virus, is used for gene transfer due to its specific characteristics:

- Foreign DNA must be packaged in viral protein for transmission.
- Inserted DNA should not disrupt virus assembly.
- CaMV lacks non-coding regions for DNA insertion.

- Non-essential genes, like gene II and gene VII, can be replaced.
- Replacement of gene II with foreign genes is feasible.
- Example: Gene II replaced with a bacterial gene for methotrexate resistance.
- Result: Turnip plants infected with modified CaMV show methotrexate resistance.

Principal of electroporation

Electroporation, also known as electro-permeabilization, involves applying a brief electrical field to living cells to create tiny pores in the cell membrane, called electro-pores. This technique is used to transfer recombinant DNA into various hosts, including bacteria, plant protoplasts, and animal cells.

Principle:

- The cell membrane's phospholipid molecules are dynamic.
- When an electric field is applied, it increases the kinetic energy of the molecules, making certain points in the membrane more permeable.
- This leads to the formation of electro-pores.
- The recombinant DNA can pass through these temporary pores before they close.

Precautions to use Murashig and skoog medium

• Ensure appropriate pH of the medium before addition of gelling agent as acidic pH will lead to decreased gelation resulting in semi solid flowing gel while alkaline pH will lead to formation of hardened gel.

• Use of Distilled water / Tissue culture grade water is recommended for media preparation as tap water or lower grade water may lead to salt precipitation and improper gelation.

• Avoid preparation of concentrated solutions, as it will lead to precipitation of salts.

Name of three plastids

Plastids Have diverse functions

• Chloroplasts – green plastids – for photosynthesis

• Chromoplasts – coloured plastids – for pigment synthesis and storage

• Gerontoplasts – control dismantling of photosynthetic apparatus during senescence

• Leucoplasts – colourless plastids – monoterpene synthesis

• Leucoplasts include amyloplasts (starch), elaioplasts (fats), proteinoplasts (proteins) and tannosomes (tannins)

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Pre –zygotic Barriers

• Those occurring prior to fertilization

• These are due to the genetic differences in different species

• Failure of pollen germination

• Slow growth of pollen tube

• Inability of the pollen to reach the ovary

• Arrest of pollen tube in style, ovary and ovule

Genetically herbicide tolerant plants/Crops

Herbicide-tolerant plants, like corn and soybeans, are developed to withstand the effects of herbicides, which are chemicals used to control weeds and minimize crop yield losses.

- Weeds compete with crops for nutrients and cause significant yield losses.
- Modern agriculture relies on herbicides, but they often have limitations in controlling all types of weeds.
- Developing herbicide-tolerant plants for broad-spectrum herbicides is a promising solution.
- Herbicides are chemicals that kill or inhibit plant growth without harming animals.
- They typically target processes unique to plants, such as photosynthesis, by inhibiting essential enzyme reactions.

- Herbicide tolerance in plants can be achieved by reducing the inhibition caused by herbicides, leading to increased plant survival and crop yield.

Technological and environmental strains of bio fertilizers

Environmental constraints

• Seasonal demand for biofertilizers

• Simultaneous cropping operations and short span of sowing/planting in a particular locality

• Soil characteristics like salinity, acidity, drought, water logging, etc.

Technological constraints

• Use of improper, less efficient strains for production.

• Lack of qualified technical personnel in production units.

• Production of poor quality inoculants without understanding the basic microbiological techniques

• Short shelf life of inoculants.

Role of auxins in plants

• Auxins stimulate cell elongation

• Influence a host of other developmental responses,

• Root initiation, vascular differentiation, tropic responses, apical dominance

• Development of auxiliary buds, flowers and fruits

• Auxins in plant tissue culture are used to induce callus from explants, and cause root and shoot morphogenesis and parthenocarpy.

How growth regulators are classified on the basis of origin?

• Hormones now referred to as growth regulators are organic compounds that have been naturally synthesized in higher plants, which influence growth and development.

• These are usually active at different sites from where they are produced and are only present and active in very small quantities.

Growth regulators, or plant hormones, are classified by their origin:

Naturally occurring: These are made by plants themselves. Examples are auxins (like IAA) and cytokinins (like zeatin).

Synthetic: These are human-made versions of plant hormones. They're used in agriculture and tissue culture.

In tissue culture, auxins and cytokinins are key players, influencing cell growth and differentiation. Other natural

regulators like gibberellins, abscisic acid, and ethylene are less important in this context.

Murashige and Skoog Medium (MS)

- Murashige and Skoog Medium (MS) was first made in 1962 for studying cytokinins in tobacco.
- It's now widely used for plant tissue culture.
- MS contains salts, vitamins, amino acids, and carbs.
- It has all the nutrients plants need, including macro and microelements.
- Potassium dihydrogen phosphate gives phosphate.
- Microelements like Boron, Manganese, etc., help plant metabolism.
- Boron is important for carbs.
- Vitamins like Thiamine, Pyridoxine, and Nicotinic Acid help plant enzymes.
- Glycine provides amino acids.

Germplasm preservation

• Germplasm is living tissue from which new plants can be grown. It can be a seed or another plant part a leaf, a piece of stem, pollen or even just a few cells that can be turned into a whole plant.

• Germplasm contains the information for a species' genetic makeup, a valuable natural resource of plant diversity.

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Why lettuce is favored over tobacco?

• Most of the plant is leaf tissue and this tissue contains the greatest number of plastids per cell

• Unlike tobacco, lettuce has no toxic alkaloids that need to be removed - low purification and downstream processing costs

• Lettuce is a relevant human foodstuff that can be consumed without cooking

3 advantage of somatic embryogenesis

• Higher propagation rate

• Suitable for suspension culture

• Artificial seed production

• Somaclonal variation

• Germplasm conservation

• Labour savings

Importance of Bio-fertilizers

(i) They increase the yield of plants by 15-35%.

(ii) Bio-fertilizers are effective even under semi-arid conditions,

(iii) Farmers can prepare the inoculum themselves,

(iv) They improve soil texture,

(v) Bio-fertilizers do not allow pathogens to flourish,

(vi) They produce vitamins and growth promoting bio-chemical's,

(vii) They are non-polluting

How crown gull disease occur?

- Crown gall disease affects plants like peaches, grapes, and roses, caused by Agrobacterium tumefaciens.
- It was postulated in 1907 that a bacterium caused crown gall tumors, later confirmed.
- tumefaciens infects wounded plant tissues, leading to tumor formation.
- Phenolic compounds released by wounded sites facilitate bacterium entry.
- Symptoms include wart-like growths above soil line, causing weakness and stunting.
- Crown gall forms when A. tumefaciens releases its Ti plasmid into plant cells.
- Ti plasmid contains T-DNA, which integrates into the plant genome, causing tumor formation.
- T-DNA carries genes for growth hormones (auxin and cytokinin) and novel metabolites used by the bacterium.
- Bacteria use opines and agropines from T-DNA for their own nutrition.

- Tumor formation interferes with plant growth due to genetic transformation.
- Various plants, especially dicots like grapes and roses, are affected by crown gall disease.

Importance of callus in tissue culture

- Callus is a mass of undifferentiated cells formed from plant tissue in tissue culture.
- It is essential for wound healing and is found in various living organisms.
- In plant tissue culture, callus is produced by culturing plant tissue on nutrient media in vitro.
- Growth hormones in the culture media stimulate callus formation and growth.
- Callus is crucial for regeneration of plants, genetic transformation experiments, mass propagation, and production of secondary metabolites.
- It serves as a platform for studying plant physiology, genetics, and biotechnology.

Inoculation

• Transfer of explant (root, stem, leaf, etc.) on to a culture medium is called inoculation.

• The inoculation is carried out under aseptic condition for which an apparatus called laminar air flow chamber is used.

• Flamed and cooled forceps are used for transfer of plant materials to different culture media kept in glasswares.

Steps tissue culture technique?

Plant tissue culture

• Plant tissue culture is a technique of growing plant cells, tissues, organs, seeds or other plant parts in a sterile environment on a nutrient medium

Basic techniques of plant tissue culture are

- Culture vessels
- Culture medium
- Sterilization
- Inoculation
- Incubation
- Induction of callus
- Morphogenesis = Organogenesis Embryogenesis
- Hardening

Composition of culture medium:

• Nutrient Medium

• Medium depends upon the type of plant tissue or cell used for culture

• Generally nutrient consist of

• Inorganic salts (both micro & macro elements) a carbon source (usually sucrose)

• Vitamins (eg. nicotinic acid, thiamine, pyridoxine

• Amino acids (eg. arginine)

• Growth regulators (eg. auxins)

• An optimum pH (5.7) is also vary important

Steps of tissue culture

Disadvantages of particles bombardment system

• Difficulty in obtaining single copy transgenic events.

• High cost of the equipment and microcarriers.

• Intracellular target is random (cytoplasm, nucleus, vacuole, plastid, etc.).

• Transfer DNA is not protected.

Sterilization of explant

Surface sterilization of explants in plant tissue culture is crucial to remove surface contaminants like bacteria and fungi. Common sterilants include household bleach and ethanol. Sterilization time varies based on tissue type.

While surface sterilization is effective, internal contamination may still occur. Mitigation involves frequent medium transfers or low antibiotic concentrations. Balancing sterilization with tissue viability is essential to prevent tissue damage.

Advantages of Somatic Embryogenesis

- Higher propagation rate
- Suitable for suspension culture
- Artificial seed production
- Somaclonal variation
- Germplasm conservation
- Labour savings

Bridge binding

Bridge binding occurs in molecular biology when two probes or primers bind to adjacent regions on a target nucleic acid molecule, forming a bridge-like structure.

It can lead to nonspecific hybridization and affect assay accuracy and specificity. Careful probe and primer design is crucial to avoid bridge binding and optimize assay performance.

Precaution of MS medium

- Ensure appropriate pH of the medium before addition of gelling agent as acidic pH will lead to decreased gelation

resulting in semi solid flowing gel while alkaline pH will lead to formation of hardened gel.

• Use of Distilled water / Tissue culture grade water is recommended for media preparation as tap water or lower grade water may lead to salt precipitation and improper gelation.

• Avoid preparation of concentrated solutions, as it will lead to precipitation of salts.

Characters of cell suspension culture

- Dispersed cell aggregates in liquid medium.
- Initiated from callus or various plant tissues.
- Mechanical initiation by grinding and filtration.
- Continuous agitation for nutrient distribution.
- Aim for rapid growth with uniform, viable cells.
- Regular subculturing, ideally weekly.
- Passage length typically 1-2 weeks with dilutions.
- Monitoring cell density before subculturing.

Benefits of agricultural biotechnology:

- Disease resistance protects crops, ensuring food security.
- Increases profitability by improving crop quality and yields.
- Enhances nutritional traits, addressing deficiencies like vitamin A.
- Herbicide-tolerant crops enable efficient weed management.

- Advances research, aiding understanding of living organisms.
- Safely controls insect pests, reducing reliance on harmful pesticides.
- Improves weed management, safeguarding crops and environment.

What is multi-partite virus?

The genome in these viruses is divided into small RNAs which may be in the same particle or different particles, e.g. brome mosaic virus (BMV). HMV contains four

RNAs divided between three particles. Plant RNA viruses, in general, are characterized by high level of gene expression, good efficiency to infect cells and spread to different tissues. But the major limitation to use them as vectors is the difficulty of joining RNA molecules in vitro.

What are protoplast? Write down the two methods of protoplast fusion?

Protoplast fusion is proposed for creating novel hybrid plants not achievable through conventional breeding methods.

Protoplasts, devoid of cell walls, can be obtained from various plant species, including important crops.

While any two plant protoplasts can be fused chemically or physically, the generation of distinct somatic hybrids faces challenges.

The main limitations lie in the ability to regenerate fused products and the sterility often observed in interspecific hybrids, rather than in protoplast production itself.

What is the need of germplasm conservation?

• Plant germplasm is genetic source material in the form of Seeds, Cultured cells Callus, Pollens .

• The in-situ /ex-situ preservation of these material is known as “Germplasm conservation”.

• Germplasm provide the raw material (genes) which the breeder uses to develop commercial crop varieties.

High lysine com

• Under the heading of ‘High-lysine com’, we have described how cereals rich in certain essential amino acids such as lysine, methionine and tryptophan can be developed by genetic engineering.

• Likewise, rice is being modified into Golden rice by Prof. Inge Potrykus and Dr. Peter Beyer.

• This is done so that vitamin A potential is maintained even after the husks are removed, a procedure adopted to allow for storage since the husks become rancid.

• This change may improve health of millions of people throughout the world.

Pathogen attack strategies

• **Necrotrophy**, in which the plant cells are killed

• **Biotrophy**, in which the plant cells remain alive

• **Hemibiotrophy**, in which the pathogen initially keeps cells alive but kills them at later stages of infection

Morphogenesis & it's types

Formation of new organs from the callus under the influence of auxin and cytokinin is called morphogenesis.

• Roots and shoots are differentiated from the callus.

• Such embryos are called somatic embryos result in the formation of young plantlet.

Types of morphogenesis

• Organogenesis

• Embryogenesis

Organogenesis

• Formation of new organs such as shoot and root is known as organogenesis.

• The development of shoot from the callus is called caulogenesis and formation of root is called rhizogenesis respectively.

Embryogenesis

• Formation of embryos (ie. bipolar structure having shoot and root) from the callus is called embryogenesis.

• These embryos arise from somatic callus tissue and are called somatic embryos or embryoids or somaclonal embryos.

Prezygotic stage of anther

The prezygotic stage of anther development involves:

- Microspore formation (microsporogenesis).
- Microspore differentiation into pollen grains (microgametogenesis).
- Release of mature pollen grains.
- Pollen germination and growth of pollen tube for fertilization.

Stress resistance mechanisms

• Avoidance mechanisms

• prevents exposure to stress

• Tolerance mechanisms

• permit the plant to withstand stress

• Acclimation

• alter their physiology in response stress

Multiplication step in tissue culture

The explant gives rise to a callus (a mass of loosely arranged cells) which is manipulated by varying sugar concentrations and the auxin (low): cytokinin (high) ratios to form multiple shoots

• The callus may be subdivided a number of times Dividing shoots

• Warmth and good light are essential

Disadvantage PEG culture

1. The DNA is susceptible for degradation and rearrangement.

2. Random integration of foreign DNA into genome may result in undesirable traits.

3. Regeneration of plants from transformed protoplasts is a difficult task.

Genetic Markers in Plant Breeding

• A genetic marker is a gene or DNA sequence with a known location on a chromosome

• It can be used to identify individuals or species.

• It can be described as a variation (which may arise due to mutation or alteration in the genomic loci) that can be observed.

• Any phenotypic difference controlled by genes, that can be used for studying recombination processes or selection of a more or less closely associated target gene

• Anything in the genome that is variable and can be used to compare individuals

• Detectable allelic variation on a chromosome it can be a phenotype, can also be a unique detectable sequence of DNA

• They are points of variation that can be used to identify individuals or species, ormay be used to associate an inherited disease with a gene through genetic linkage with nearby but possibly unidentified or uncharacterised genes.

• Examples include single nucleotide polymorphisms (SNPs) and minisatellites.

Stages of anther

Pre-mitotic stage:

• Anthers at this stage have microspores which have just completed the first meiotic division and the pollens are immature, uninucleate and starch-free.

• Anthers of Hordeum vulgare and Hyocyamus at this stage are suitable for culture. According to Nitsch (’72) and Sunderland (’71) anthers with uninucleate pollens are suitable for culture.

• b. Mitotic stage:

• In some plants, anthers at first pollen division stage are most suitable for culture, as observed in Nicotiana tabacum and Datura innoxia.

• c. Post-mitotic stage:

• Early bi-cellular stage of pollen development is most suitable for culture in Atropa belladonna and Nicotiana sp. etc.

Why do we need to check the colonies in a transgenic medium.

Checking colonies in a transgenic medium is necessary to:

- Select transgenic cells resistant to selective agents.
- Confirm transgene integration into the genome.
- Assess transformation efficiency.
- Ensure quality control and detect contamination.

Direct Organogenesis

• The production of direct buds or shoots from a tissue with no intervening callus stage is termed direct organogenesis.

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• Plants have been propagated by direct organogenesis for improved multiplication rates, production of transgenic plants, and—most importantly—for clonal propagation.

• Typically, indirect organogenesis is more important for transgenic plant production.

By studying agriculture biotechnology in details?? How would you describe that agricultural biotechnology is beneficial for ourselves.

Agricultural biotechnology benefits humanity by:

- Increasing crop yield and quality.
- Promoting sustainable farming practices.
- Enhancing crop nutrition.
- Controlling plant diseases.
- Adapting to climate change.
- Producing biopharmaceuticals.
- Conserving the environment and biodiversity.

What are agrobacterium mediated transformation.

Agrobacterium-mediated transformation uses Agrobacterium tumefaciens, a soil bacterium, to transfer foreign DNA into plant cells. A. tumefaciens naturally infects plants, causing crown gall disease, and is harnessed for genetic engineering due to its ability to transfer DNA into plant genomes.

This technique revolutionizes plant biotechnology by introducing desired traits into crops, such as pest resistance and improved nutritional content.

Somatic hybridization

Somatic hybridization, also known as protoplast fusion, involves the fusion of protoplasts from different plant species to generate hybrids. Protoplasts are plant cells lacking a cell wall but containing a plasma membrane and other cellular components. Despite the absence of a cell wall, protoplasts retain full cellular functionality.

By fusing protoplasts from different species, somatic hybridization allows for the creation of hybrids with combined traits from the parent species. This technique has applications in plant breeding and genetic engineering for developing novel crop varieties with desirable characteristics

Stages of Micropropagation

Micropropagation, also known as tissue culture propagation, involves the propagation of plants through the growth of small plant parts in a laboratory setting. The process typically consists of several stages:

Step 0 - Pre-culture: Initial selection and growth of common plants under controlled conditions for approximately 3 months.

Stage I - Establishment: Initiation and establishment of cultures in a suitable medium. Explants (plant parts) such as organs, shoot tips, or axillary shoots are selected, surface sterilized, and placed in culture to initiate growth.

Stage II - Multiplication: Main activity of micropropagation occurs here, with explants multiplying rapidly in a defined culture medium. Shoots or explant embryos are multiplied.

Stage III - Rooting: Transfer of shoots to a medium conducive to rapid root development. In vitro rooting is preferred for handling a large number of species, but sometimes shoots are planted directly in soil.

Stage IV - Acclimatization: Establishment of seedlings in soil. Seedlings from the laboratory (Stage III) are transferred to a greenhouse environment. In some cases, Stage III is omitted, and nonrooted shoots from Stage II are directly planted in pots or suitable compost mix.

Steps of development of plants

Plant development progresses through several stages:

- Germination: Seed absorbs water and embryo emerges.
- Seedling Establishment: Cotyledons emerge, roots and shoots develop.
- Vegetative Growth: Plant elongates, leaves expand, branches may form.

- Reproductive Initiation: Flower buds develop in response to environmental cues.
- Flowering: Flowers form, undergo pollination, fertilization, and seed formation.
- Fruit and Seed Development: Ovary develops into fruit, enclosing seeds.
- Seed Dispersal: Fruits and seeds dispersed by various means.
- Seed Dormancy: Some seeds enter a dormant stage.
- Seed Germination: Dormant seeds sprout when conditions are favorable, restarting the cycle.

Protoplast culture

Isolated protoplasts cultured in medium to regenerate cell wall and form callus.

Optimal conditions: suitable density, balanced auxin-to-cytokinin ratio, glucose/sucrose, osmoprotectants, temperature (20-28°C), pH (5.5-5.9), and presence of casein hydrolysate, BAP, and NAA.

Protoplasts form new cell walls, increase in size and organelles, and undergo cell division. First division occurs in 2-7 days, leading to microcolonies within 1-3 weeks.

Zoology Legends

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Remember me in you Precious Prayer

Jazak Allah

Team Zoology LEGENDS